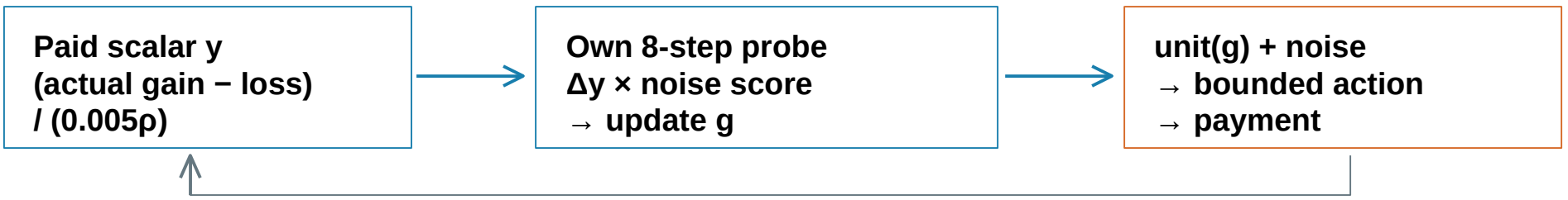
Scalar feedback supports acquired action with limited revision

Reward reversal and a changed actuator test different capabilities.

A Local scalar receipts, carried state and a preset actuator



Physical K transports material and internal state; it is not a policy input.

Visible: local scalar, carried state, own clock/RNG and completed-payment flag.
No gradient, field position, donor displacement or actuator truth is supplied.

B Reward reversal succeeds in only one of two conditions

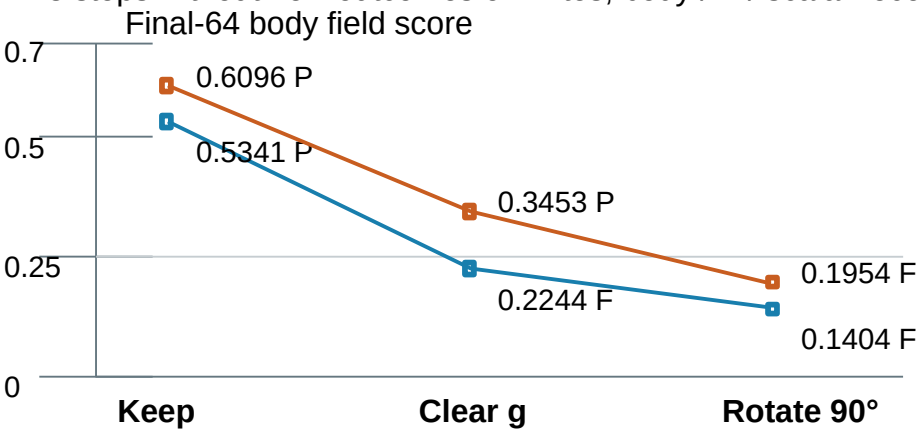
Evaluation: final-64 body field score | P / F = original full gate

Method	1–320 Acquire	321–640 Reverse	641–768 No feedback
Response estimate	**C0 +0.6355 P** **C1 +0.6327 P**	**C0 +0.1468 F** **C1 +0.2798 P**	**C0 +0.5341 P** **C1 +0.6096 P**
Reward delta	C0 +0.4588 P C1 +0.4625 P	C0 -0.2411 F C1 -0.1850 F	C0 +0.2160 F C1 +0.1773 F
Context delta	C0 +0.4555 P C1 +0.4586 P	C0 -0.5243 F C1 -0.5163 F	C0 -0.4279 F C1 -0.2654 F
Fixed run/tumble	C0 +0.2586 P C1 +0.1608 F	C0 -0.0274 F C1 -0.0193 F	C0 +0.0721 F C1 +0.0474 F
Random	C0 +0.0657 F C1 -0.0085 F	C0 -0.0511 F C1 +0.0276 F	C0 -0.0869 F C1 +0.0370 F

Gate: score > .25; positive mass fraction ≥ .60; M ≥ 12; connected fraction ≥ .75.
C0 / C1 are two conditions. Success after 640 does not repair failure during reversal.

C Same actual-640 parents: stored content is useful, but not exclusive

128 steps without new outcomes or writes; body / E / scalar records / clock / RNG matched.



C0 parent → three branches
C1 parent → three branches

Pass: keep 2/2; clear 1/2; rotate 0/2.

Clear still explores and transports.
A favorable body position can persist.
Lines connect matched branches,
not successive time points.
Grey line: score threshold only.

D Changed actuation: a cheaper fixed feedback control also succeeds

Actual-320 parents; same reward meaning. Fixed run/tumble receives the same initial heading.

Branch	321–640 Map changed	641–768 No feedback	Movement cost / EU
Continue updates	C0 +0.8899 P C1 +0.8685 P	C0 +0.8562 P C1 +0.7657 P	C0 1.4221 C1 1.3829
Frozen estimate	C0 -0.2329 F C1 -0.2939 F	C0 -0.2532 F C1 -0.2866 F	C0 1.4873 C1 1.4872
Fixed run/tumble	C0 +0.7126 P C1 +0.7076 P	C0 +0.7199 P C1 +0.7208 P	C0 1.2037 C1 1.2101

Frozen writers still receive physical outcomes. Movement totals cover steps 321–768.
No tissue birth; dense feedback. 10 life0 + 10 dependent branches, not 20 independent repeats.
Actual outcomes / basal / movement recorded; computation, communication and sensing hardware unpriced.